AGRIMITRA

AI-POWERED SMART SOIL MONITORING FOR SMALL FARMERS

KNOW YOUR SOIL. GROW SMARTER. EARN MORE.

BUILD FOR AUTOMATE INDIA 2026

TEAM NAME: CODE CRUSHERS

THE PROBLEM

"Raju spent ₹12,000 on fertilizers this season. His crop still failed. His soil was already toxic — he just had no way to know."
On the slide:

86% of India's 140 million farmers are small & marginal — with zero access to real-time soil data.

Why current testing fails:

Costs ₹500–₹2,000 per test
Takes 3 to 21 days for results
Tested once every 3–5 years, if at all
Available only near urban or semi-urban labs

What this causes:

- 20–40% crop yield loss due to poor soil decisions
- High fertilizer costs and wasted inputs
- Long-term soil damage from over-fertilization

"The soil is speaking. Farmers just don't have the tools to listen."

Four Soil Vital Signs (24/7 Monitoring) :

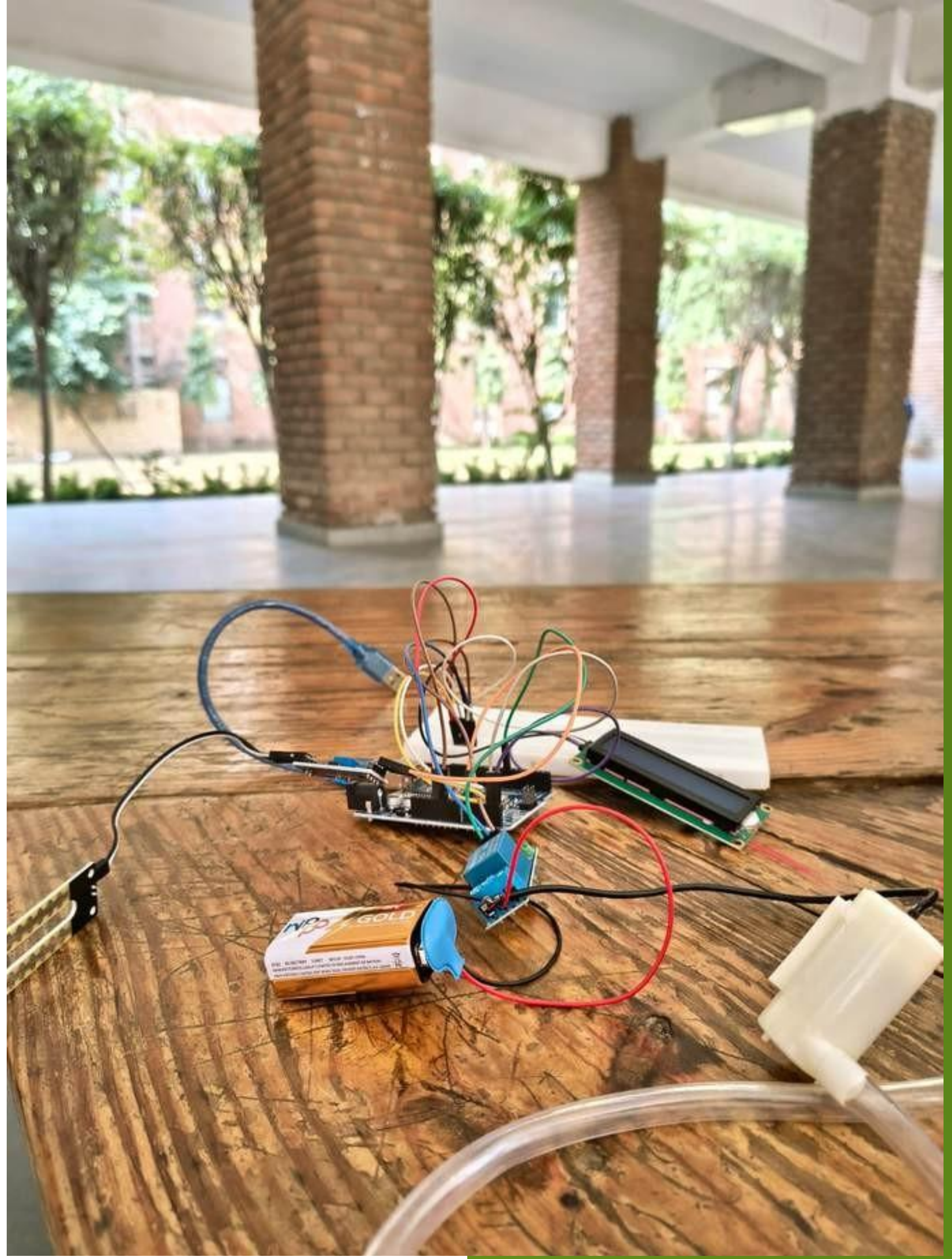
Soil Moisture Sensor – Measures water content and guides proper irrigation.

MQ-Series Gas Sensor – Detects harmful gases indicating soil stress or fertilizer imbalance.

Temperature Sensor – Monitors soil temperature affecting nutrient absorption.

Humidity Sensor – Tracks environmental conditions influencing crop health and disease risk.

IMPLEMENTATION



01. Sensor Layer (Field Monitoring)

• Soil Moisture Sensor measures water content in the root zone
• MQ-Series Gas Sensor detects harmful soil gases and chemical imbalance
• DHT11 Sensor monitors temperature and humidity affecting crop growth

02 Processing Layer (Smart Controller)

• ESP8266 / NodeMCU collects real-time sensor data
• On-device AI algorithm processes the data
• Generates a Soil Health Score (0–100) indicating soil condition

03 Alert & Decision Layer

• Detects abnormal soil conditions automatically
• Sends alerts via LED / buzzer / SMS notification
• Helps farmers take quick irrigation and fertilizer decisions

04 Farmer & Financial Interface

- Farmers view live Soil Health Score via display or SMS
- Data history generates a Farmer Soil Report
- Can support loan eligibility, subsidies, and financial credit

small demo:[https://agrimitraweb.netlify .app/](https://agrimitraweb.netlify.app/)

Agrimitra converts complex soil sensor data into a simple Soil Health Score (0–100), helping farmers easily understand soil conditions and take timely action.

Score	Status	What the Farmer Does
80-100	Excellent	Continue current practices
60-79	Good	Minor adjustments needed
40-59	Moderate	Soil treatment recommended
20-39	Poor	Urgent correction required
0-19	Critical	Immediate intervention needed

Innovation 2: Farmer Credit System:

Agrimitra transforms soil performance into Soil Credit Points, generating a Farmer Soil Report that can support access to loans, subsidies, and financial services.

Tier	Score	Benefit Unlocked
Platinum	90-100	Priority loan processing + full subsidy access
Gold	70-89	Reduced interest rate on agricultural loans
Silver	50-69	Standard loan eligibility unlocked
Bronze	30-49	Advisory support and guidance access

IMPACT AND SDG

- Provides early detection of soil problems, helping prevent crop losses
- Reduces fertilizer misuse by 30–50%, lowering farming costs
- Replaces guesswork with a simple 0–100 Soil Health Score
- Improves crop productivity and soil sustainability

SDG	Goal	Our Contribution
SDG 2	Zero Hunger	Improves soil health and crop productivity at the grassroots, strengthening local food security for smallholder farmers.
SDG 8	Decent Work & Economic Growth	Connects sustainable farming to formal finance through Soil Credit Points and Farmer Soil Reports that unlock loans, lower interest rates, and higher incomes.
SDG 12	Responsible Consumption & Production	Reduces over-application of fertilizers and promotes precision, data-driven resource use on farmland.
SDG 15	Life on Land	Prevents long-term soil degradation and supports sustainable land-management practices through early soil-health alerts and interventions.

- Generates Soil Credit Points and Farmer Soil Reports
- Supports access to loans, subsidies, and insurance for farmers
- Promotes data-driven and climate-smart farming practices

SCALABILITY AND FUTURE SCOPE

Scalability

- Collaboration with Farmer Producer Organizations (FPOs) for large-scale farmer adoption
- Support from Agricultural Extension Officers for field deployment and monitoring
- Partnerships with seed and fertilizer companies for wider distribution

Future Scope

- Integration of pH, NPK, and EC sensors for advanced soil nutrient analysis
- Smart irrigation and drone-based spraying linked to Soil Health Score
- AI platform combining soil data, weather data, and satellite imagery for climate-smart farming

WHY IT MAKES US DIFFERENT?

• Affordable Solution – Device cost around ₹800, making it accessible for small farmers.
• Real-Time Monitoring – Provides continuous soil health data instead of delayed lab reports.
• Simple Soil Health Score – Converts complex sensor data into an easy 0–100 score that farmers can understand instantly.
• Early Warning System – Detects soil stress and harmful gas levels before visible crop damage occurs.
• Farmer Credit System – Generates Soil Credit Points and Farmer Soil Reports to support access to loans and subsidies.

THANK-YOU

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